Transfer Learning

Deep Learning

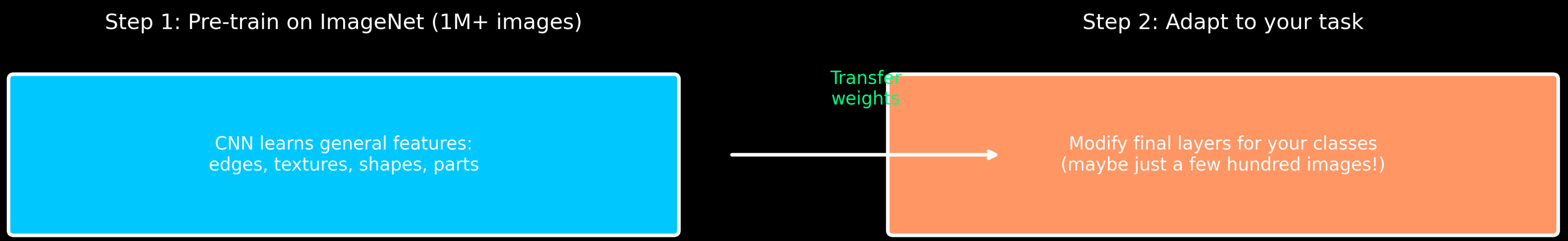
Daniel E. Acuna
Associate Professor, University of Colorado Boulder

Contents of This Video

In this video, we will cover:

- What is transfer learning?
- Why pre-trained features transfer
- Feature extraction vs. fine-tuning
- How to use pre-trained models
- When transfer learning works best

The Transfer Learning Idea



✓ Works with small datasets ✓ Trains in minutes ✓ Often beats training from scratch

Why Features Transfer

CNN layers learn hierarchical features:

Layer	Features	Transferability
Early	Edges, colors	Very general (always useful)
Middle	Textures, patterns	Fairly general
Late	Object parts	Somewhat task-specific
Final	Full objects	Task-specific

Early/middle layers transfer well to almost any image task

Two Approaches

Feature Extraction:

- Freeze pre-trained layers
- Only train new classifier head
- Fast, works with tiny datasets
- Best when your data is similar to ImageNet

Fine-Tuning:

- Unfreeze some/all layers
- Train entire network (small LR)
- Better if you have more data
- Adapts features to your domain
- Use small learning rate!

Transfer Learning Steps

Step 1: Load a pre-trained model

- Use a model trained on a large dataset (e.g., ImageNet)
- Remove the original classification layers

Step 2: Freeze the base model

- Keep pre-trained weights fixed initially

Step 3: Add new classification head

- Global average pooling to reduce spatial dimensions
- One or two dense layers
- Final softmax for your classes

Step 4: Train only the new layers

- Training is fast because most weights are frozen

When to Use What?

Your Data	Amount	Approach
Similar to ImageNet	Small (100s)	Feature extraction
Similar to ImageNet	Large (10K+)	Fine-tune top layers
Different from ImageNet	Small	Feature extraction + augmentation
Different from ImageNet	Large	Fine-tune whole network

Start with feature extraction, try fine-tuning if needed

Transfer Learning Impact

Pre-trained models are available for many architectures:

- ResNet, VGG, EfficientNet
- Many others

Works for:

- Medical imaging
- Satellite imagery
- Industrial inspection
- Any image classification task

This is how most real-world computer vision is done!

What We've Covered

In this video, we've learned:

- Transfer learning: use pre-trained models for new tasks
- Early CNN layers learn transferable features
- Feature extraction: freeze pre-trained weights, train new head
- Fine-tuning: unfreeze layers and train with small LR
- Always start with transfer learning for image tasks